UNIVERSITY PHYSICS REFERENCE SERIES • VOLUME 2

Newton's Laws & Classical Dynamics

Inertial Reference Frames, Vector Formulations of Newton's Three Laws, Systematic Free-Body Isolation, Coulomb Friction, Multi-Body Pulley Constraints, and Centripetal Orbit Dynamics

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Synthesized From:

• University Physics Lecture Series (Chapter 2: Newton's Laws & Multi-Body Systems)
• R. Shankar — *Fundamentals of Physics: Mechanics, Relativity, and Thermodynamics* (Chapters 3 & 4)
• Halliday, Resnick, & Walker — *Fundamentals of Physics* (Force & Motion Principles)

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Dynamics is the branch of classical mechanics that investigates the causal relationship between the forces exerted on a physical body and the resulting change in its state of motion. While kinematics provides the mathematical language to describe trajectories, velocity vectors, and accelerations, Newtonian dynamics provides the physical principles explaining *why* bodies accelerate.

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Part I: Foundations of Classical Dynamics

Chapter 1: Reference Frames & The Principle of Inertia

The cornerstone of Newtonian mechanics is the concept of a **reference frame**—a coordinate system equipped with a synchronized time-keeping mechanism used by an observer to measure the positions, velocities, and accelerations of physical objects.

> **▶ Definition: Inertial Reference Frame**
>
> An **inertial reference frame** is one in which Newton's First Law holds true without modification: any body free from net external forces remains at rest or maintains a constant velocity vector along a rectilinear path:
>
> $$\mathbf{F}_{net} = 0 \iff \mathbf{a} = \frac{d\mathbf{v}}{dt} = 0 \iff \mathbf{v} = \text{const}$$

1.1 Galilean Relativity & Invariance

If a reference frame S is inertial, any other frame S' translating relative to S at a constant velocity $\mathbf{u}$ is also an inertial frame. Under the Galilean transformation:

$$\mathbf{r}' = \mathbf{r} - \mathbf{u}t, \quad \mathbf{v}' = \mathbf{v} - \mathbf{u}, \quad \mathbf{a}' = \frac{d\mathbf{v}'}{dt} = \frac{d(\mathbf{v} - \mathbf{u})}{dt} = \mathbf{a}$$

Because accelerations are identical in all mutually non-accelerating frames ($\mathbf{a}' = \mathbf{a}$), Newton's laws are invariant under Galilean transformations. No mechanical experiment conducted entirely within a closed laboratory can detect the uniform rectilinear speed of the laboratory.

1.2 Non-Inertial Frames & Inertial (Fictitious) Forces

When an observer measures motion from a frame S_{acc} that accelerates with linear acceleration $\mathbf{a}_{frame}$ relative to an inertial frame, Newton's Second Law in its standard form appears to fail. A particle with zero real physical interaction accelerates at $\mathbf{a}_{rel} = -\mathbf{a}_{frame}$.

To retain the formal structure of $\mathbf{F} = m\mathbf{a}$ within such an accelerating frame, one must introduce an **inertial (fictitious) force**:

$$\mathbf{F}_{inertial} = -m\,\mathbf{a}_{frame} \implies m\,\mathbf{a}_{rel} = \mathbf{F}_{real} + \mathbf{F}_{inertial} = \mathbf{F}_{real} - m\,\mathbf{a}_{frame}$$

Prominent examples include the d'Alembert force in accelerating vehicles, the centrifugal force $\mathbf{F}_{cf} = m\omega^2\mathbf{r}$ in rotating frames, and the velocity-dependent Coriolis force $\mathbf{F}_{Cor} = -2m(\boldsymbol{\omega} \times \mathbf{v}_{rel})$.

> **⚠️ Methodological Rule: Avoiding Fictitious Forces**
>
> Throughout this handbook, all standard Free-Body Diagrams (FBDs) are strictly formulated from an **inertial ground frame**. In an inertial frame, centrifugal forces DO NOT EXIST; centripetal acceleration is the kinematic consequence of real inward forces (gravity, normal force, tension, friction).

Chapter 2: Newton's Three Laws of Motion

2.1 Newton's First Law: The Law of Inertia

"Every body perseveres in its state of being at rest or of moving uniformly straight forward, except insofar as it is compelled to change its state by forces impressed."

The First Law is not merely a special case of the Second Law with **a** = 0. Its primary philosophical role is to **postulate the existence of inertial reference frames**. Inertia is the intrinsic property of matter that resists any deviation from uniform rectilinear motion, quantified by the scalar quantity known as **inertial mass** *m*.

2.2 Newton's Second Law: Vector Momentum Formulation

The net external force acting on a body equals the time rate of change of its linear momentum **p** = *m***v**:

$$\mathbf{F}_{\text{net}} = \sum \mathbf{F}_{\text{ext}} = \frac{d\mathbf{p}}{dt} = \frac{d(m\mathbf{v})}{dt}$$

For a classical system with constant invariant mass ($dm/dt = 0$), this simplifies to the familiar fundamental equation of dynamics:

$$\sum \mathbf{F} = m\,\mathbf{a} = m\,\frac{d^2\mathbf{r}}{dt^2}$$

In three-dimensional Cartesian space, this vector equation resolves into three mutually independent scalar differential equations:

$$\sum F_x = m\,a_x, \qquad \sum F_y = m\,a_y, \qquad \sum F_z = m\,a_z$$

> **▶ Shankar's Insight: The Two Halves of Newton's Second Law**
>
> As emphasized by R. Shankar in *Fundamentals of Physics*, the equation **F** = *m***a** consists of two conceptually distinct components:
>
> 1. **The Left-Hand Side (F):** Specifies the *force law* governing the physical interactions dictated by the environment (e.g., Hooke's law $-kx$, Universal gravitation $-GMM/r^2$, Coulomb friction μN).
> 2. **The Right-Hand Side (*m*a):** Dictates the *kinematic response* of a particle of inertial mass *m* to that force via second-order differential calculus.

2.3 Newton's Third Law: Mutual Action and Reaction

"To every action there is always opposed an equal reaction; or the mutual actions of two bodies upon each other are always equal and directed to contrary parts."

$$\mathbf{F}_{A \text{ on } B} = -\mathbf{F}_{B \text{ on } A}$$

> **⚠️ Essential Distinction: Action-Reaction Pairs vs. Balanced Equilibrium Forces**
>
> A frequent misconception among physics students is confusing Newton's Third Law pairs with balanced forces under Newton's First or Second Law:

Property	Newton's 3rd Law Pair	Balanced Forces (Equilibrium)
Bodies Involved	Acts on **two different** bodies (Body A and Body B)	Acts on the **same single** body
Physical Origin	Always identical interaction type (both gravitational, or both normal)	Can be completely different physical origins (e.g., Gravity vs Normal)
Net Force Cancellation	**Never cancel** on an isolated Free-Body Diagram	Sum vectorially to zero on the single body: $\sum \mathbf{F} = 0$
Example: Book on Table	Earth pulls book down ⟺ Book pulls Earth up	Table pushes book up (**N**) balances Earth pulls book down (**W**)

Part II: Free-Body Diagrams & Fundamental Forces

Chapter 3: Systematic Free-Body Diagram (FBD) Methodology

The Free-Body Diagram is the definitive graphical and analytical tool of Newtonian mechanics. It represents an isolated system of interest stripped of its surroundings, displaying all external forces exerted *on* the system by the environment.

▶ The 4-Step Systematic FBD Protocol

1. **Isolate the Target Body:** Replace the physical object with an idealized point particle at its center of mass, isolated from all adjacent surfaces, cords, or external bodies.
2. **Identify Field (Non-Contact) Forces:** Draw long-range forces acting through space—primarily the downward gravitational weight vector $\mathbf{W} = m\mathbf{g}$ directed towards Earth's center.
3. **Identify Contact Forces at the Physical Boundaries:** Examine every point of mechanical contact on the body's perimeter:
   - **Perpendicular Solid Contacts:** Normal force **N** acting perpendicular to the contact surface into the body.
   - **Parallel Solid Contacts:** Friction force **f** acting parallel to the surface, opposing relative motion or impending slippage.
   - **Tension Attachments:** Cords or cables pulling away from the body along the line of the string with tension **T**.
   - **Elastic Contacts:** Spring forces $\mathbf{F}_{sp} = -k\Delta\mathbf{x}$ along the axis of extension/compression.
4. **Establish Natural Coordinate Axes:** Align coordinate axes such that one axis coincides with the anticipated direction of acceleration **a**. All orthogonal axes will then satisfy equilibrium: $\sum F_{perp} = 0$.

Chapter 4: Contact & Field Forces

4.1 The Normal Force: A Self-Adjusting Contact Force of Constraint

The normal force **N** is an electromagnetic contact force generated by electron cloud repulsion between microscopic surface asperities. It is a *constraint force*: its magnitude is not set by an independent constitutive law, but adjusts dynamically to satisfy the kinematic constraint of rigid non-penetration (relative acceleration perpendicular to the boundary must be zero).

Consequently, ***N* is not automatically equal to *mg***. It depends directly on applied vertical forces, angles of inclination, and vertical accelerations:

Horizontal flat table (rest): $N = mg$
Pulling upward at angle ϕ: $N = mg - P \sin \phi$
Pushing downward at angle ϕ: $N = mg + P \sin \phi$
Inclined plane of angle θ: $N = mg \cos \theta$

4.2 Tension in Ideal Strings

An **ideal string** is defined as perfectly flexible (zero bending resistance), completely unstretchable (inextensible: L = const), and massless ($m_{string} = 0$).

Applying Newton's Second Law to an infinitesimal segment of a massless string yields:

$$\sum F = T_1 + T_2 = m_{segment}\, a = 0 \implies T_1 = T_2 = T$$

Hence, **tension is uniform throughout an ideal string**, even when accelerating. It transmits force around ideal (frictionless, massless) pulleys without alteration in magnitude, merely redirecting the force vector along the cord's path.

4.3 Weight vs. Apparent Weight in Accelerating Frames

While true gravitational weight is the body force **W** = m**g**, **apparent weight** W_{app} is the sensation of heaviness registered by a spring scale or floor supporting the observer. It equals the magnitude of the supporting normal force **N**:

$$\sum F_y = N - mg = m\, a_y \implies W_{app} = N = m\,(g + a_y) = m\, g_{eff}$$

- **Elevator Accelerating Upward (+a_y):** $N = m(g + a) > mg$ (Feels heavier).
- **Elevator Accelerating Downward (-a_y):** $N = m(g - a) < mg$ (Feels lighter).
- **Free-Fall State ($a_y = -g$):** $N = m(g - g) = 0$ (Apparent weightlessness).

Part III: Friction & Inclined Plane Dynamics

Chapter 5: The Coulomb-Amontons Laws of Dry Friction

Friction is the resistive tangential contact force that opposes the relative lateral sliding of two bodies in physical contact. At the microscopic scale, macroscopic surfaces touch only at microscopic peaks called *asperities*. The localized pressure at these microscopic contact points is high enough to induce atomic cold-welding.

5.1 Static Friction (f_s) and the Maximum Threshold

When an external tangential force $\mathbf{F}_{\text{app}}$ is applied to a stationary body, static friction self-adjusts to precisely balance the applied force, keeping the body in static equilibrium:

$$f_s = F_{\text{app}} \quad (\text{for } 0 \le F_{\text{app}} \le f_{s,\max})$$

The junction can only resist shear stresses up to a critical maximum static threshold $f_{s,\max}$, which is proportional to the normal force pressing the surfaces together:

$$f_s \le \mu_s N, \quad f_{s,\max} = \mu_s N$$

where μ_s is the dimensionless **coefficient of static friction**.

5.2 Kinetic Friction (f_k) and Stick-Slip Transitions

Once the applied tangential force exceeds $f_{s,\max}$, the microscopic cold-welds rupture, and relative sliding motion begins. The resistive force drops abruptly to the **kinetic friction** force:

$$f_k = \mu_k N$$

For virtually all standard material interfaces, $\mu_k < \mu_s$. This discontinuity explains the familiar *stick-slip* phenomenon (e.g., squealing tires, violin strings, earthquake fault slips).

Chapter 6: Inclined Plane Mechanics & Vector Decomposition

When analyzing motion on an inclined plane with inclination angle θ relative to the horizontal, standard horizontal and vertical Cartesian axes force acceleration components onto both axes. The optimal strategy is to **rotate the coordinate system**:

- **Parallel Axis ($+x_{\parallel}$):** Aligned down the plane (direction of sliding motion).
- **Perpendicular Axis ($+y_{\perp}$):** Aligned perpendicular to the plane (pointing outward).

Resolving the vertical gravitational weight vector $\mathbf{W} = m\mathbf{g}$ onto these rotated axes:

$$W_{\parallel} = mg \sin\theta \quad \textbf{(tangential component pulling down the incline)}$$
$$W_{\perp} = mg \cos\theta \quad \textbf{(normal component pressing into the incline)}$$

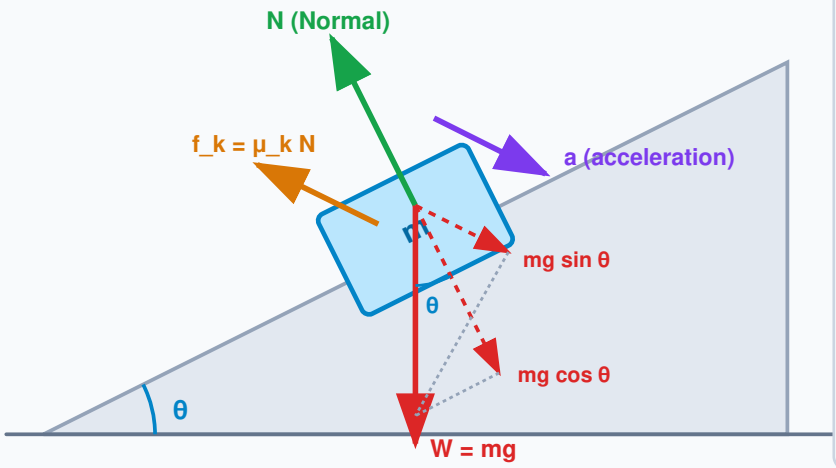
Applying Newton's Second Law along both axes:

$$\sum F_{\perp} = N - mg\cos\theta = 0 \implies N = mg\cos\theta$$
$$\sum F_{\parallel} = mg\sin\theta - f_k = ma \implies ma = mg\sin\theta - \mu_k(mg\cos\theta)$$

$$a = g\,(\sin\theta - \mu_k \cos\theta) \quad \text{(sliding down an incline)}$$

Figure 2.1: Inclined Plane Dynamics & Rotated Coordinate Decomposition

Free-body resolution along the natural incline axes (x∥ down-plane, y⊥ perpendicular)



Rotated Coordinate Axes



Equations of Motion:

$\Sigma F_y = N - mg\cos\theta = 0$

$\Rightarrow N = mg\cos\theta$

$\Sigma F_x = mg\sin\theta - f_k = ma$

$f_k = \mu_k mg\cos\theta$

Resulting Acceleration:

$a = g(\sin\theta - \mu_k \cos\theta)$

Figure 2.1: Free-Body Diagram of a block on an inclined plane showing rotated coordinate decomposition along x∥ and y⊥ axes.

Part IV: Coupled Systems & Pulley Kinematics

Chapter 7: Kinematic Constraints in Pulley Systems

When multiple bodies are mechanically coupled through cords and pulleys, their accelerations are not independent; they are linked by geometric **kinematic constraint equations**.

7.1 Inextensible String Constraint

For an ideal string of total length L wrapped around fixed pulleys connecting two masses with positions x_1 and x_2:

$$x_1 + x_2 + \text{const} = L \implies \frac{d^2x_1}{dt^2} + \frac{d^2x_2}{dt^2} = 0 \implies a_1 + a_2 = 0 \implies |a_1| = |a_2| = a$$

7.2 The Classical Atwood Machine

Two masses m_1 and m_2 (with $m_2 > m_1$) hang vertically from opposite ends of a light cord over a frictionless pulley:

$$\textbf{Body 1 (accelerating upward):} \quad T - m_1 g = m_1 a$$
$$\textbf{Body 2 (accelerating downward):} \quad m_2 g - T = m_2 a$$

Adding the two equations eliminates the internal tension force T immediately:

$$a = \frac{m_2 - m_1}{m_1 + m_2}\, g, \qquad T = \frac{2\, m_1\, m_2}{m_1 + m_2}\, g$$

7.3 The Modified Atwood Machine

A block of mass m_1 on a horizontal surface connected via a light cord over a corner pulley to a hanging mass m_2 (Figure 2.2).

Figure 2.2: Modified Atwood Machine & Multi-Body System Decomposition

Connected particle constraints with string tension transmission and isolated free-body diagrams

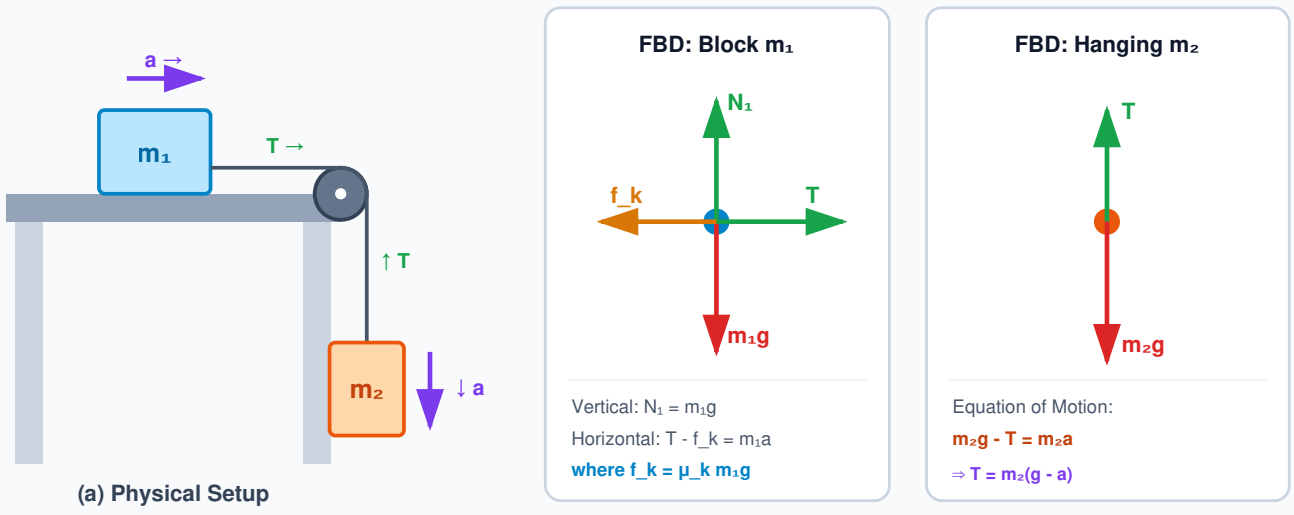


Figure 2.2: Modified Atwood Machine setup with separate free-body diagrams for horizontal slider m1 and hanging mass m2.

Chapter 8: Multi-Block Stacks & Relative Motion Thresholds

In multi-block stacked systems (e.g., Block 1 resting atop Block 2), the only horizontal force capable of accelerating Block 1 is the **static friction force** exerted by the surface of Block 2:

$$f_{s,\max} = \mu_s m_1 g \implies a_{1,\max} = \frac{f_{s,\max}}{m_1} = \mu_s g$$

If the combined system acceleration exceeds $a_{\max} = \mu_s g$, static friction fails, and Block 1 slips relative to Block 2.

Part V: Centripetal Dynamics & Circular Motion

Chapter 9: Centripetal Acceleration & Net Radial Forces

When a particle moves along a curved path of radius r at instantaneous speed v, its velocity vector continuously changes direction even if its scalar speed remains constant. This directional change demands a kinematic acceleration directed radially inward toward the local center of curvature:

$$\mathbf{a}_c = -\frac{v^2}{r}\,\hat{\mathbf{r}} = -\omega^2 r\,\hat{\mathbf{r}} = -\frac{4\pi^2 r}{T^2}\,\hat{\mathbf{r}}$$

▶ **Fundamental Principle: Centripetal Force is a Net Force Requirement**

Centripetal force is **not an independent physical force**. There is no such thing as an autonomous "centripetal interaction." Rather, "centripetal" is a *geometric job description*:

$$\sum F_r = m\,a_c = m\,\frac{v^2}{r}$$

Whatever real physical forces point radially inward (tension, gravity, normal force, friction) must sum to provide the required quantity $m\,v^2 / r$.

Chapter 10: Curved Trajectories & Banked Roadways

10.1 Flat (Unbanked) Curves

On a flat horizontal curve of radius R, the only horizontal force available to supply centripetal acceleration is the **static friction force** between the tires and the road:

$$f_s = m\,\frac{v^2}{R} \le f_{s,\max} = \mu_s\,mg \implies v \le v_{\max} = \sqrt{\mu_s\,g\,R}$$

10.2 Ideal (Frictionless) Banked Curves

By tilting (banking) the roadway by an angle θ, the normal force **N** tilts toward the center of curvature. Its horizontal component $N \sin\theta$ provides the requisite centripetal acceleration without relying on friction:

$$\textbf{Vertical Equilibrium:}\quad N\cos\theta = mg \implies N = \frac{mg}{\cos\theta}$$

$$\textbf{Horizontal Dynamics:}\quad N\sin\theta = m\,\frac{v^2}{R} \implies \tan\theta = \frac{v^2}{g\,R} \implies v_{\text{opt}} = \sqrt{R\,g\tan\theta}$$

Figure 2.3: Banked Curve Centripetal Dynamics & Lateral Friction Limits

Resolution of normal force N and static friction f_s into horizontal (radial) and vertical components

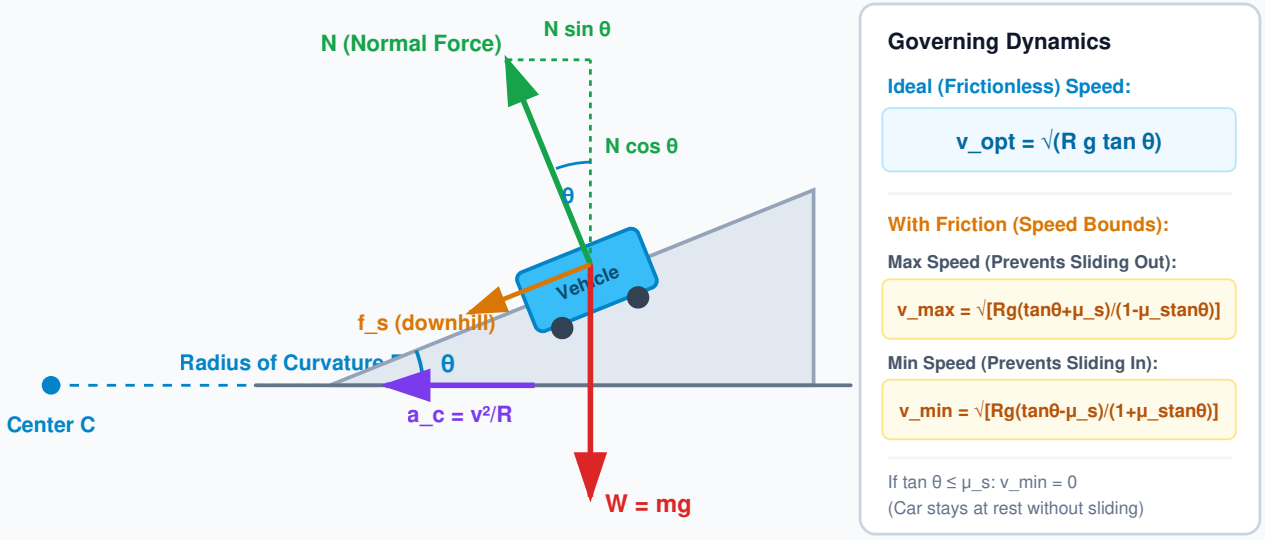


Figure 2.3: Banked roadway force resolution showing normal force N, gravity mg, static friction fs, and centripetal acceleration ac.

10.3 The Conical Pendulum

A bob of mass m suspended from a cord of length L tracing a horizontal circle of radius $r = L \sin\theta$ at constant speed v:

$$\text{Vertical: } T\cos\theta = mg, \quad \text{Horizontal: } T\sin\theta = m\frac{v^2}{r} \implies \tan\theta = \frac{v^2}{rg}$$

$$T_{\text{period}} = \frac{2\pi r}{v} = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

Chapter 11: Vertical Circular Motion & Loop-the-Loop Dynamics

In a vertical circle of radius R, gravity acts parallel to the path at the sides and perpendicular to the path at the top and bottom. Consequently, the speed v is non-uniform, dictated by the conservation of mechanical energy:

$$\frac{1}{2}m v_{\text{bot}}^2 = \frac{1}{2}m v_{\text{top}}^2 + mg(2R) \implies v_{\text{bot}}^2 = v_{\text{top}}^2 + 4gR$$

11.1 Apex Contact Threshold & Nadir Dynamics

At the top of the loop, both gravity and track normal force point radially downward:

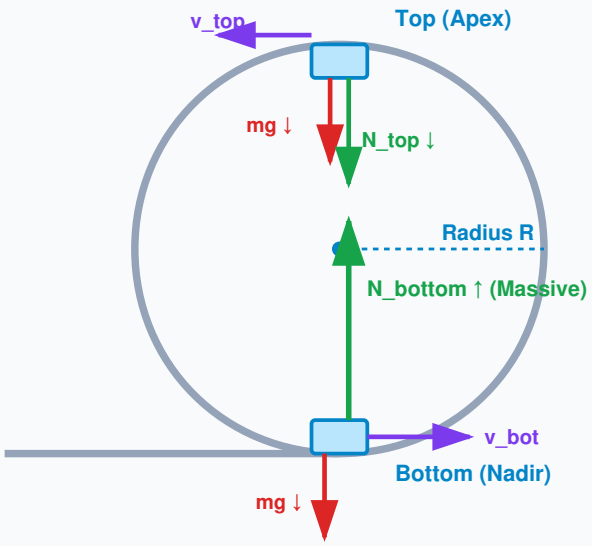
$$\sum F_r = N_{\text{top}} + mg = m\frac{v_{\text{top}}^2}{R} \implies N_{\text{top}} = m\frac{v_{\text{top}}^2}{R} - mg \geq 0 \implies v_{\text{top,crit}} = \sqrt{gR}$$

At the lowest point, the normal force points upward, while gravity points downward:

$$N_{\text{bot}} = mg + m\frac{v_{\text{bot}}^2}{R} = N_{\text{top}} + 6mg$$

Figure 2.4: Vertical Circular Loop Dynamics & Critical Speed Limits

Force analysis at the apex and nadir of a vertical loop under gravity



Loop Boundary Conditions

1. Condition to Maintain Contact at Top:

$\Sigma F_r = N_{top} + mg = m v_{top}^2 / R$

To not lose contact: $N_{top} \geq 0$

$$v_{top,crit} = \sqrt{gR}$$

2. Dynamic Load at Bottom:

$\Sigma F_r = N_{bottom} - mg = m v_{bot}^2 / R$

$\Rightarrow N_{bottom} = mg + m v_{bot}^2 / R$

⚡ The Universal 6-mg Spread:

By mechanical energy conservation:

$\frac{1}{2}mv_{bot}^2 = \frac{1}{2}mv_{top}^2 + mg(2R)$

$$\Delta N = N_{bottom} - N_{top} = 6mg$$

Figure 2.4: Dynamics of vertical circular motion highlighting the apex contact threshold and the universal 6-mg normal force spread.

11.2 Artificial Gravity in Rotating Spacecraft

In deep space where true gravity is negligible, artificial gravity is simulated on the inner perimeter of a cylindrical habitat of radius R rotating at angular speed ω. The floor provides a normal force directed inward:

$$N = m\,a_c = m\,\omega^2 R = m\,g_{\text{earth}} \implies \omega = \sqrt{\frac{g}{R}}, \quad v = \sqrt{(g\,R)}$$

Part VI: ⚡ Fast-Track Hand Calculation Shortcuts

⚡ FAST-TRACK 1: The "Whole-System" Shortcut for Coupled Masses

5 SECONDS

Standard Method (Slow): Write separate FBDs for all n bodies, set up simultaneous 2nd-law linear systems, and algebraically substitute to eliminate intermediate cord tensions T_1, T_2, etc.

⚡ Fast Hand Calculation Shortcut: Treat the entire connected assembly as a single collective super-body of mass $M_{total} = \sum m_i$. All internal string tensions cancel out by Newton's Third Law!

$$a = \frac{\sum F_{external,\ driving} - \sum F_{external,\ opposing}}{M_{total}}$$

Example: Block m_1 = 3 kg on frictionless table pulled by hanging m_2 = 2 kg.

$F_{driving} = m_2 g = 2(9.8) = 19.6$ N. $M_{total} = 3 + 2 = 5$ kg.

$a = 19.6 / 5 = 3.92$ m/s² instantly in your head!

⚡ FAST-TRACK 2: Atwood Acceleration & Tension Ratio Rules

MENTAL MATH

Standard Method (Slow): Solve $T - m_1 g = m_1 a$ and $m_2 g - T = m_2 a$ through multi-step substitution.

⚡ Fast Hand Calculation Shortcut: Express acceleration as the difference-over-sum mass ratio times g, and tension as twice the reduced mass μ_{red} times g:

$$a = \left(\frac{\Delta m}{M_{tot}}\right) g = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g, \qquad T = 2\,\mu_{red}\, g = \frac{2\, m_1\, m_2}{m_1 + m_2}\, g$$

Example: m_1 = 4 kg, m_2 = 6 kg.

$\Delta m / M_{tot} = (6 - 4)/(10) = 0.20 \implies$ **$a = 0.20(9.8) = 1.96$ m/s²**.

$T = 2(24/10)(9.8) = 4.8(9.8) =$ **47.04 N**.

⚡ FAST-TRACK 3: Instant Angle of Repose Rule

ZERO CALCULATION

Standard Method (Slow): Balance $mg \sin\theta$ with $\mu_s\, mg \cos\theta$, divide both sides by m, divide by g, and simplify trigonometric terms.

⚡ Fast Hand Calculation Shortcut: The critical angle of repose θ_c at which any block begins to slide on an incline depends strictly on friction, completely independent of mass or gravity:

$$\tan\theta_c = \mu_s \iff \theta_c = \arctan(\mu_s)$$

Rule: If $\mu_s = 0.577$ ($1/\sqrt{3}$), $\theta_c = 30°$ instantly! If tilt $\theta < \theta_c$, block remains stationary.

⚡ FAST-TRACK 4: Effective Gravity (g_{eff}) for Accelerating Frames

1 STEP

Standard Method (Slow): Re-derive equations of motion inside non-inertial frames using coordinate transformation tensors.

⚡ **Fast Hand Calculation Shortcut: Replace standard g with effective gravity vector $g_{eff} = g - a_{frame}$:**

$$\text{Vertical elevator: } g_{eff} = g \pm a \implies N = m\, g_{eff}$$

$$\text{Horizontal accelerating cart: } g_{eff} = \sqrt{(g^2 + a^2)}, \quad \text{pendulum tilt } \tan \beta = a/g$$

⚡ FAST-TRACK 5: Ideal Banked Speed Shortcut

FORMULA TRICK

⚡ **Fast Hand Calculation Shortcut: An ideal banked curve of angle θ and a conical pendulum of angle θ share the exact same governing dynamic speed equation:**

$$v_{opt} = \sqrt{(R\, g \tan \theta)} \iff \tan \theta = \frac{v^2}{R\, g}$$

⚡ FAST-TRACK 6: The Universal 6-mg Law for Vertical Circular Loops

INVARIANT LAW

⚡ **Fast Hand Calculation Shortcut: For any frictionless vertical circular loop under gravity, the difference between the bottom normal force and the top normal force is universally constant:**

$$\Delta N = N_{bottom} - N_{top} = 6\, mg \quad \text{(Exactly 6 times the body's weight!)}$$

Part VII: Step-by-Step Worked Problems

Problem 1: Block on Rough Incline with Angled Applied Force

Inclined Planes & Friction

Problem Statement: A crate of mass m = 12.0 kg rests on an inclined plane making an angle θ = 30.0° with the horizontal. The coefficient of kinetic friction between the crate and the incline is μ_k = 0.250. A pulling force **P** is applied to the crate directed at an angle ϕ = 20.0° above the incline surface. Determine the magnitude of the pulling force P required to drag the crate up the incline at a constant speed v = 1.50 m/s.

1. GIVEN PARAMETERS & COORDINATE ALIGNMENT

- Mass: m = 12.0 kg, Incline angle: θ = 30.0°, Pull angle relative to incline: ϕ = 20.0°
- Kinetic friction coefficient: μ_k = 0.250, Acceleration: a = 0 (constant speed), g = 9.80 m/s^2
- Rotated axes: $+x_{\parallel}$ uphill, $+y_{\perp}$ perpendicular outward from incline.

2. SYMBOLIC ALGEBRAIC DERIVATION

Perpendicular axis ($+y_{\perp}$): $\sum F_{\perp} = N + P\sin\phi - mg\cos\theta = 0 \Longrightarrow N = mg\cos\theta - P\sin\phi$.

Kinetic friction: $f_k = \mu_k N = \mu_k (mg\cos\theta - P\sin\phi)$.

Parallel axis ($+x_{\parallel}$) with a = 0: $\sum F_{\parallel} = P\cos\phi - mg\sin\theta - f_k = 0$:

$$P = mg \cdot \frac{\sin\theta + \mu_k \cos\theta}{\cos\phi + \mu_k \sin\phi}$$

3. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

- mg = (12.0 kg)(9.80 m/s^2) = 117.6 N
- Numerator: sin(30.0°) + (0.250)cos(30.0°) = 0.5000 + 0.2165 = **0.7165**
- Denominator: cos(20.0°) + (0.250)sin(20.0°) = 0.9397 + 0.0855 = **1.0252**
- Pulling force: P = (117.6 N) × (0.7165 / 1.0252) = **82.19 N ≈ 82.2 N**

Physical Sanity & Boundary Check:
1. Normal force check: N = 117.6 cos 30° - 82.19 sin 20° = 101.84 - 28.11 = 73.73 N > 0 (no lift-off).
2. Limit ϕ = 0 (pulling parallel): P = 117.6(0.7165 / 1.0) = 84.26 N. Pulling at ϕ = 20° requires less force (82.19 N < 84.26 N) because the upward component relieves normal force and friction!

Problem 2: Coupled Double-Mass System on Incline with Friction

Modified Atwood Machine

Problem Statement: A block of mass m_1 = 8.00 kg rests on an inclined surface with slope θ = 37.0°. The coefficients of friction between m_1 and the incline are μ_s = 0.350 and μ_k = 0.200. Block m_1 is connected by a light, inextensible cord passing over a frictionless, massless pulley to a vertically hanging block m_2 = 12.00 kg. Determine: (a) whether the system moves from rest, (b) the magnitude of acceleration a, and (c) the tension T in the cord.

1. EQUILIBRIUM TENDENCY & MOTION CHECK

• Downward pull on m_2: $m_2 g$ = (12.00)(9.80) = **117.60 N**.

• Downhill gravitational component on m_1: $m_1 g \sin\theta$ = (8.00)(9.80) sin(37.0°) = **47.18 N**.

• Impending motion: m_2 descends, m_1 slides uphill since 117.60 N > 47.18 N.

• Normal force on m_1: $N_1 = m_1 g \cos\theta$ = (78.40) cos(37.0°) = **62.61 N**.

• Maximum static friction: $f_{s,\max} = \mu_s N_1$ = (0.350)(62.61 N) = **21.91 N**.

• Net driving force: $F_{\text{net,drive}}$ = 117.60 N - 47.18 N = **70.42 N** > $f_{s,\max}$ (21.91 N) $\Longrightarrow$ **System accelerates!**

2. SYMBOLIC ALGEBRAIC DERIVATION (DYNAMIC STATE)

Kinetic friction opposes uphill sliding: $f_k = \mu_k m_1 g \cos\theta$ = (0.200)(62.61 N) = 12.52 N.

Applying Fast-Track 1 (Whole-System Newton's 2nd Law):

$$a = \frac{m_2 g - m_1 g \sin\theta - \mu_k m_1 g \cos\theta}{m_1 + m_2}, \qquad T = m_2 (g - a)$$

3. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

• Net unbalanced force: F_{net} = 117.60 N - 47.18 N - 12.52 N = **57.90 N**

• System acceleration: a = 57.90 N / (8.00 + 12.00 kg) = **2.895 m/s² ≈ 2.90 m/s²**

• Cord tension: T = (12.00 kg)(9.80 m/s² - 2.895 m/s²) = **82.86 N ≈ 82.9 N**

Physical Sanity & Boundary Check:

1. Independent tension check via Block 1: $T = m_1 a + m_1 g \sin\theta + f_k$ = (8.00)(2.895) + 47.18 + 12.52 = **82.86 N** (identical).
2. Bounds check: $0 < a < g$, and $m_1 g \sin\theta < T < m_2 g$ strictly satisfied.

Problem 3: Conical Pendulum Dynamics & Cable Rupture Limit

Circular Motion & Tensile Limits

Problem Statement: A small steel bob of mass m = 1.80 kg is suspended from a light steel cable of length L = 2.50 m. The bob is set into uniform circular motion in a horizontal plane, forming a conical pendulum. The cable has a maximum tensile rupture rating of T_{max} = 45.0 N. Determine: (a) the maximum cone half-angle θ_{max} before cable snapping, (b) the maximum orbital speed v_{max}, and (c) the minimum orbital period T_{period} at this threshold.

1. GIVEN PARAMETERS & PHYSICAL GEOMETRY

• Cable length: L = 2.50 m, Bob mass: m = 1.80 kg, Max tension: T_{max} = 45.0 N, g = 9.80 m/s^2

• Orbital radius: $r = L \sin\theta$, Vertical altitude: $h = L \cos\theta$.

2. SYMBOLIC ALGEBRAIC DERIVATION

Vertical equilibrium: $\sum F_z = T\cos\theta - mg = 0 \implies \cos\theta = mg / T$.

Horizontal radial dynamic equation: $\sum F_r = T\sin\theta = m v^2 / r$.

$$\boldsymbol{\theta_{max} = \arccos\left(\frac{mg}{T_{max}}\right), \quad v = \sqrt{(r\, g \tan\theta)}, \quad T_{period} = 2\pi\sqrt{\frac{L\cos\theta}{g}}}$$

3. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

• Bob weight: mg = (1.80 kg)(9.80 m/s^2) = **17.64 N**

• Max angle: $\cos(\theta_{max})$ = 17.64 / 45.0 = 0.3920 $\implies$ $\boldsymbol{\theta_{max}}$ **= 66.92° ≈ 66.9°**

• Radius: r_{max} = (2.50 m) sin(66.92°) = **2.30 m**

• Speed: v_{max} = √[(2.30 m)(9.80 m/s^2) tan(66.92°)] = **7.27 m/s**

• Period: T_{period} = 2π √[(2.50 × 0.3920) / 9.80] = 2π √(0.100) = **1.987 s ≈ 1.99 s**

Physical Sanity & Boundary Check:

1. Independent check: $v = 2\pi r / T_{period}$ = 2π(2.30) / 1.987 = **7.27 m/s** (exact agreement).
2. Small angle limit: as $\theta \to 0$, $T_{period} \to 2\pi\sqrt{(L/g)}$ = 3.17 s (recovers simple pendulum period).

Problem 4: Banked Highway Curve with Friction Bounds

Centripetal Force & Road Design

Problem Statement: An expressway turn with radius of curvature R = 150.0 m is banked at an angle θ = 15.0° above the horizontal. (a) Calculate the ideal cruising speed v_{opt} for which no lateral friction is required. (b) Under icy conditions with static friction coefficient μ_s = 0.300, determine the safe speed envelope $[v_{min}, v_{max}]$ within which an automobile can negotiate the curve without skidding.

1. GIVEN PARAMETERS & GEOMETRIC RATIOS

• Radius: R = 150.0 m, Banking angle: θ = 15.0° (tan 15.0° = 0.26795), μ_s = 0.300, g = 9.80 m/s².

2. SYMBOLIC ALGEBRAIC DERIVATION

$$v_{opt} = \sqrt{(R\, g \tan\theta)}, \qquad v_{max} = \sqrt{\frac{R\, g\, (\tan\theta + \mu_s)}{1 - \mu_s \tan\theta}}$$

If $\tan\theta \le \mu_s$, static friction prevents sliding down at zero velocity, so $\boldsymbol{v_{min} = 0}$.

3. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

• Ideal speed: v_{opt} = √[(150.0)(9.80)(0.26795)] = **19.85 m/s = 71.5 km/h**

• Minimum speed check: tan(15.0°) = 0.268 < μ_s = 0.300 ⟹ $\boldsymbol{v_{min}}$ **= 0 m/s (0 km/h)**.

• Maximum speed: v_{max} = √[(1470)(0.56795 / 0.91962)] = √(1470 × 0.61759) = **30.13 m/s = 108.5 km/h**

Physical Sanity & Boundary Check:
1. Safe Speed Envelope: **0 km/h ≤ v ≤ 108.5 km/h**. The vehicle has complete stability from dead stop up to freeway speeds.
2. Check limit μ_s = 0: v_{max} = √[1470(0.26795)] = 19.85 m/s, smoothly collapsing to v_{opt}.

Problem 5: Stacked Two-Block System on Frictionless Floor

Multi-Body Friction Coupling

Problem Statement: Block 1 of mass m_1 = 3.00 kg is stacked on top of Block 2 of mass m_2 = 7.00 kg. The coefficient of static friction between the two blocks is μ_s = 0.400. Block 2 rests on a frictionless horizontal floor (μ_{floor} = 0). Determine: (a) the maximum horizontal force F_1 that can be applied to the **top** block without relative slippage, and (b) the maximum horizontal force F_2 that can be applied to the **bottom** block without relative slippage. Explain the physical origin of the difference.

1. GIVEN PARAMETERS & INTERFACIAL NORMAL FORCE

• Top mass: m_1 = 3.00 kg, Bottom mass: m_2 = 7.00 kg, Total mass: M = 10.00 kg, μ_s = 0.400.

• Normal force at block interface: $N_{12} = m_1 g$ = (3.00 kg)(9.80 m/s²) = **29.40 N**.

• Maximum static friction force available: $f_{s,max} = \mu_s N_{12}$ = (0.400)(29.40 N) = **11.76 N**.

2. SYMBOLIC ALGEBRAIC DERIVATION

Case (a): Force on Top Block (m_1): Block 2 is accelerated solely by friction: $a_{max,(a)} = f_{s,max} / m_2 = \mu_s m_1 g / m_2$.

Case (b): Force on Bottom Block (m_2): Block 1 is accelerated solely by friction: $a_{max,(b)} = f_{s,max} / m_1 = \mu_s g$.

$$F_{1,max} = (m_1 + m_2) \frac{\mu_s m_1 g}{m_2}, \qquad F_{2,max} = (m_1 + m_2) \mu_s g$$

3. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

• **Case (a):** $a_{max,(a)}$ = 11.76 N / 7.00 kg = 1.680 m/s² $\Longrightarrow$ $F_{1,max}$ = (10.00 kg)(1.680 m/s²) = **16.80 N**.

• **Case (b):** $a_{max,(b)}$ = (0.400)(9.80 m/s²) = 3.920 m/s² $\Longrightarrow$ $F_{2,max}$ = (10.00 kg)(3.920 m/s²) = **39.20 N**.

Physical Sanity & Comparative Analysis:
Notice that $F_{2,max}$ (39.2 N) is more than **2.3 times larger** than $F_{1,max}$ (16.8 N)!
Physical Rationale: When pulling the top block, the friction interface must drag the heavy 7 kg bottom block, bottlenecking the maximum acceleration to 1.68 m/s². Conversely, when pulling the bottom block, friction only needs to drag the light 3 kg top block, allowing more than double the acceleration (3.92 m/s²) before slippage occurs.

Part VIII: Comprehensive Master Dynamics Formula Sheet

This reference sheet consolidates all governing formulas for classical Newtonian dynamics, incorporating required equations from the university physics examination curriculum:

Domain / Topic	Governing Formula	Operational Conditions
Newton's 2nd Law	$\sum \mathbf{F} = \frac{d\mathbf{p}}{dt} = m\,\mathbf{a}$	Inertial frame, constant mass m
Newton's 3rd Law	$\mathbf{F}_{A \text{ on } B} = -\mathbf{F}_{B \text{ on } A}$	Mutual action-reaction pairs on different bodies
Static Friction	$f_s \le \mu_s N, \quad f_{s,\max} = \mu_s N$	Static equilibrium; impending motion threshold
Kinetic Friction	$f_k = \mu_k N$	Active relative surface sliding
Angle of Repose	$\tan\theta_c = \mu_s$	Critical angle for motion onset on incline
Incline Acceleration	$a = g\,(\sin\theta \mp \mu_k \cos\theta)$	- for sliding down, + for pushed up
Atwood Machine	$a = \frac{m_2 - m_1}{m_1 + m_2}\,g, \quad T = \frac{2\,m_1\,m_2}{m_1 + m_2}\,g$	Massless, inextensible string, ideal pulley
Apparent Weight	$N = m\,(g \pm a_y) = m\,g_{\text{eff}}$	+ for upward acceleration, - for downward
Centripetal Acceleration	$a_c = \frac{v^2}{r} = \omega^2 r = \frac{4\pi^2 r}{T^2}$	Uniform/non-uniform circular motion (radial)
Net Centripetal Force	$\sum F_r = m\,\frac{v^2}{r} = m\,\omega^2\,r$	Sum of real physical radial force components
Conical Pendulum	$\tan\theta = \frac{v^2}{r\,g}, \quad T_{\text{period}} = 2\pi\sqrt{\frac{L\cos\theta}{g}}$	Bob orbiting in horizontal circle of radius $L\sin\theta$
Ideal Banked Road	$\tan\theta = \frac{v^2}{R\,g} \implies v_{\text{opt}} = \sqrt{(R\,g\tan\theta)}$	No lateral tire friction required
Banked with Friction	$v_{\max} = \sqrt{\frac{Rg(\tan\theta + \mu_s)}{1 - \mu_s\tan\theta}}$	Upper speed threshold preventing outward skidding
Vertical Loop (Apex)	$v_{\text{top,crit}} = \sqrt{(g\,R)}, \quad N_{\text{top}} = 0$	Minimum speed to maintain contact at loop zenith
Vertical Loop (Spread)	$\Delta N = N_{\text{bottom}} - N_{\text{top}} = 6\,mg$	Universal for frictionless vertical circular motion
Artificial Gravity	$g_{\text{eff}} = \omega^2 R \implies \omega = \sqrt{\frac{g}{R}}$	Simulated terrestrial gravity in rotating habitat

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